

3.3 Tool Schema

The tool schema determines the surface form of admissible tool calls. Taking `v1.0` as the base schema, we provide two environment shift scenarios of varying degrees, where the stronger shift (`v2.0`) extends the milder one (`v1.1`) through additional updates to the tool invocation protocols. Specifically, `v1.0` is the base schema itself, consisting of 13 tools defined in ALFWorld with intuitive verb-style names such as `Go`, `Take`, `Heat`, `Slice`, and `Look`. The second, `v1.1`, applies paraphrasing: a semantics-preserving renaming of every tool name and parameter key from `v1.0` into more descriptive forms (e.g., `Go` $\rightarrow$ `NavigateTo`, with its parameter `receptacle` $\rightarrow$ `destination`), leaving the cardinality of the tool set unchanged. The third, `v2.0`, additionally applies grouping by structural and functional similarity: building on `v1.1`, tools that share structural or functional roles are consolidated into a smaller, higher-level set, reducing the cardinality from 13 to 5. Each consolidated tool exposes its sub-operations through a discrete action parameter. For example, `NavigateTo`, `OpenContainer`, and `CloseContainer` are merged into a single `ReceptacleControl` tool, whose action parameter takes one of three values: `navigate_to`, `open_container`, or `close_container`. Table 2 provides an illustrative example of the three versions, with full details in Appendix B.2.

Across the three versions, both p and TE are updated accordingly, with details deferred to Appendix B.2. Tool names, parameter keys, and descriptions in p are revised to match the active schema, while the overall structure of p is pre-